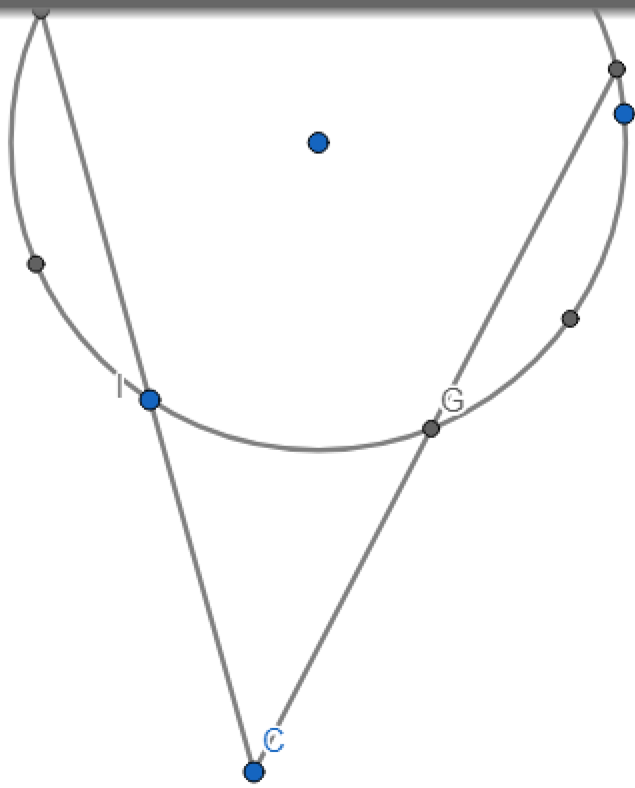
Euclidean Geometry In Mathematical Olympiads

Lemma 1.44 (Three Tangents)

Chapter 1 fgm



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Mar 16, 2019, 6:36 PM • 1 #8

Solution

$BFEC$ is cyclic, so $\measuredangle FBC = \measuredangle FEC = \measuredangle FEA$ and $\measuredangle ECB = \measuredangle EFB = \measuredangle EFA$, then let $AD \parallel BC \implies \measuredangle DAB = -\measuredangle ABC = \measuredangle AEF \implies AD$ is tangent to (AEF), now let $M' \in BC$ such that $M'E, M'F$ tangent to (AEF), so $\measuredangle M'EF = \measuredangle EAF$ and $\measuredangle M'FE = \measuredangle FAE \implies \measuredangle ECM' = \measuredangle M'EC$ and $\measuredangle FBM = \measuredangle M'FB$, then $BM' = FM' = EM' = CM' \implies M = M'$, so we are done.

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May 14, 2019, 5:05 PM • 2 #9

@above
I believe your proof is incorrect as you assume the tangents from E and F intersect on BC (by letting $M' \in BC$).

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May 15, 2019, 1:40 AM • 2 #10

oops you're right